TASK 2

Synthesis & Gate-Level Simulation

1. Aim :

To perform synthesis of a Verilog design and verify its functionality using gate-level simulation.

2. Objective

To convert RTL design into gate-level netlist

To understand synthesis process

To verify that synthesized design behaves same as RTL

3. Tools Used

- EDA Playground (for RTL simulation)
- Conceptual understanding of Cadence/Synopsys tools
- Icarus Verilog (for simulation)

4. Theory

- **Synthesis** is the process of converting RTL (Verilog code) into a **gate-level netlist** using logic gates like AND, OR, NOT, Flip-Flops.
- **Gate-Level Simulation (GLS)** is done after synthesis to ensure that the synthesized circuit behaves the same as the original RTL design.
-

5. Design Used

- The **4-bit counter** is used for synthesis and gate-level verification.

6. Gate-Level Verilog Code (Synthesized Design)

```
module counter_gate(clk, reset, out);

input clk, reset;

output reg [3:0] out;


always @(posedge clk or posedge reset)

begin

   if(reset)
```

```
        out <= 4'b0000;
    else
        out <= out + 1;
end
endmodule
```

7. Testbench for Gate-Level Simulation

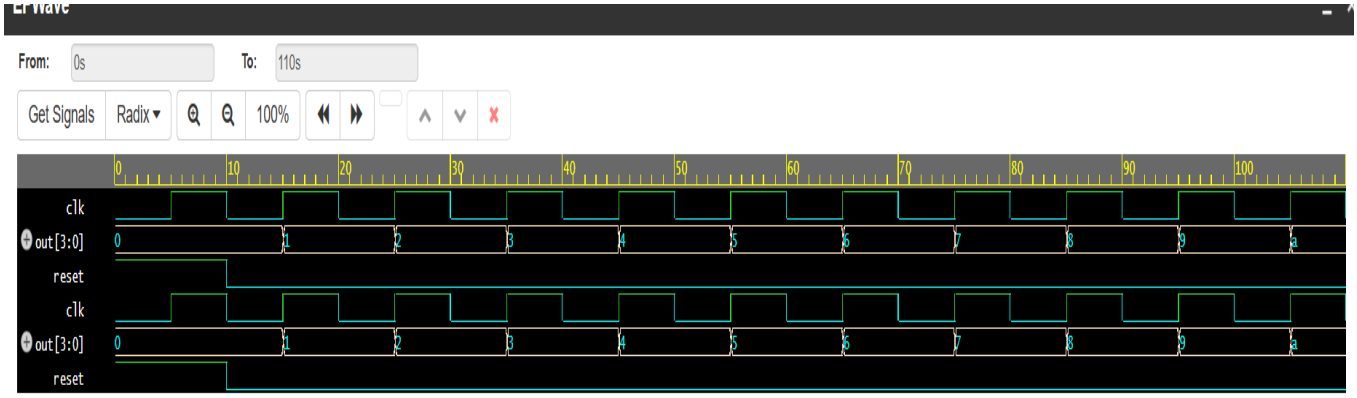
```
module tb;
reg clk, reset;
wire [3:0] out;

counter_gate uut(clk, reset, out);

always #5 clk = ~clk;

initial begin
    $dumpfile("dump.vcd");
    $dumpvars(0, tb);

    clk = 0;
    reset = 1;

    #10 reset = 0;
    #100 $finish;
end
endmodule
```

8. Simulation Procedure

1. Load gate-level code and testbench
2. Run simulation
3. Open waveform viewer

4. Observe output behavior

9. Output Waveform



10. Result

The gate-level simulation output matches the RTL simulation

The counter increments correctly on clock edges

Reset initializes the output to zero

Thus, synthesis and gate-level verification are successful.

11. Conclusion

The RTL design was successfully synthesized into a gate-level representation. The gate-level simulation confirmed that the synthesized circuit maintains the same functionality as the original design.